Statistical Signal Processing
Lecture 02:
Random signals in time-domain: correlation functions

Literature and Problems: See course schedule

Agenda (preliminary):

Random signal, X(t)

- **from random variable X to random signal X(t)**
- **simultaneous pdf**
- **moments give no information about time-variation of signal**

One signal X(t): Autocorrelation function, $R_{xx}(\tau)$

- **general definition: autocorrelation function and autocovariance function**
- **WSS (Weak Sense Stationary) definition**
- **intuitive explanation**
- **examples/demos**
- **general properties for $R_{xx}(\tau)$**
- **examples of theoretical $R_{xx}(\tau)$ calculation → problems 2.3 and 2.4**
- **stationarity: WSS and SSS (Strict Sense Stationary)**
- **estimation of $R_{xx}(\tau)$: ensemble averaging, E[ ], versus time-averaging, A[ ]**
- **the concept of ergodicity**
- **MATLAB calculation of $R_{xx}(k)$, *xcorr* and options**

Two signals X(t), Y(t): Crosscorrelation function, $R_{xy}(\tau)$

- **WSS definition**
- **intuitive explanation**
- **example/demo**
- **general properties for $R_{xy}(\tau)$**
- **alternative definitions of $R_{xy}(\tau)$**
- **time-averaging estimation of $R_{xy}(\tau)$**
- **MATLAB calculation of $R_{xy}(k)$, *xcorr* and options**

Correlation matrices (discrete-time)

- **length N signals X(n) and Y(n)**
- **autocorrelation matrix R_X: Toeplitz and symmetric**
- **covariance matrix $\sum_X$: symmetric but not Toeplitz**
- **crosscorrelation matrix R_{XY}: Toeplitz but not symmetric**